

Big Data Project

Software:

NYC Taxi/Uber Trip Analysis

The Goal: Find the "Gold Mine" neighborhoods where drivers make the most tips or experience the longest delays.

- **Data Source:** The [NYC TLC Trip Record Data](#) (Parquet files). These datasets are massive (millions of rows).
- **Tech Stack:** * **Apache Spark (PySpark):** Running on a local machine or a free Databricks Community Edition cluster.
 - **SQL:** Using SparkSQL to query the data.
- **The "Big Data" Hook:** Students learn to handle **Parquet files** (columnar storage) rather than simple CSVs, which is a staple of industry data lakes.

Network

Network Intrusion Detection (Security)

The Goal: Identify suspicious patterns in network traffic that might indicate a DDoS attack or a breach.

- **Data Source:** [Kaggle](#) dataset.
- **Tech Stack:**
 - **Hadoop/HDFS:** To store the large log files.
 - **Hive:** To run SQL-like queries on the logs.